Table 1: Execution Time of SPEC CPU 2017

Test case	LLVM-O3	Magellan	MLGO	LLM-Inliner
600.perlbench_s	206.89	–	–	191.76 ↓ 7.31%
602.gcc_s	327.60	331.21 ↑ 1.10%	357.84 ↑ 9.23%	310.22 ↓ 5.30%
605.mcf_s	520.17	493.35 ↓ 5.16%	496.67 ↓ 4.52%	495.49 ↓ 4.74%
619.lbm_s	519.44	526.91 ↑ 1.44%	533.29 ↑ 2.67%	507.78 ↓ 2.24%
620.omnetpp_s	266.53	275.30 ↑ 3.29%	291.63 ↑ 9.42%	258.97 ↓ 2.84%
625.x264_s	81.02	84.88 ↑ 4.76%	84.34 ↑ 4.10%	81.86 ↑ 1.04%
631.deepsjeng_s	299.54	299.06 ↓ 0.16%	309.99 ↑ 3.49%	279.86 ↓ 6.57%
638.imagick_s	220.65	216.48 ↓ 1.89%	215.02 ↓ 2.55%	209.08 ↓ 5.24%
641.leela_s	293.48	299.24 ↑ 1.96%	302.87 ↑ 3.20%	280.79 ↓ 4.32%
644.nab_s	189.38	193.42 ↑ 2.13%	201.11 ↑ 6.19%	189.91 ↑ 0.28%
657.xz_s	398.25	402.56 ↑ 1.08%	378.01 ↓ 5.08%	369.81 ↓ 7.14%
Geomean (s)	270.98	280.68 ↑ 3.58%	285.34 ↑ 5.30%	**259.94** ↓ **4.07%**

, ,

Table 2: Execution Time of CBench

Test case	LLVM-O3	Magellan	MLGO	LLM-Inliner
automotive_bitcount	1118	1058 ↓ 5.37%	1988 ↑ 77.82%	998 ↓ 10.73%
automotive_qsort1	1672	1560 ↓ 6.70%	1572 ↓ 5.98%	1616 ↓ 3.35%
automotive_susan_c	1620	1749 ↑ 7.96%	1936 ↑ 19.51%	1769 ↑ 9.20%
automotive_susan_e	1500	1830 ↑ 22.00%	1492 ↓ 0.53%	1472 ↓ 1.87%
automotive_susan_s	1389	1384 ↓ 0.36%	1334 ↓ 3.96%	1361 ↓ 2.02%
bzip2d	2729	2572 ↓ 5.75%	2819 ↑ 3.30%	2563 ↓ 6.08%
bzip2e	2616	2572 ↓ 1.68%	2290 ↓ 12.46%	2338 ↓ 10.63%
consumer_jpeg_c	2729	2776 ↑ 1.72%	2736 ↑ 0.26%	2736 ↑ 0.26%
consumer_jpeg_d	4212	4132 ↓ 1.90%	4133 ↓ 1.88%	4405 ↑ 4.58%
consumer_lame	1512	1517 ↑ 0.33%	1541 ↑ 1.92%	1532 ↑ 1.32%
consumer_tiff2bw	4085	4200 ↑ 2.82%	4135 ↑ 1.22%	3875 ↓ 5.14%
consumer_tiff2rgba	4734	5372 ↑ 13.48%	5112 ↑ 7.98%	5069 ↑ 7.08%
consumer_tiffdither	4159	4086 ↓ 1.75%	4070 ↓ 2.14%	4019 ↓ 3.37%
consumer_tiffmedian	3018	2908 ↓ 3.64%	3080 ↑ 2.05%	2717 ↓ 9.97%
network_dijkstra	161	153 ↓ 4.97%	165 ↑ 2.48%	168 ↑ 4.35%
network_patricia	1085	1029 ↓ 5.16%	1023 ↓ 5.71%	992 ↓ 8.57%
office_ghostscript	1785	1759 ↓ 1.46%	1880 ↑ 5.32%	1721 ↓ 3.59%
office_rsynth	1505	1393 ↓ 7.44%	1609 ↑ 6.91%	1357 ↓ 9.83%
office_stringsearch1	1264	1288 ↑ 1.90%	–	1079 ↓ 14.64%
security_blowfish_d	3921	4040 ↑ 3.03%	4072 ↑ 3.85%	3913 ↓ 0.20%
security_blowfish_e	4131	4213 ↑ 1.98%	4148 ↑ 0.41%	3932 ↓ 4.82%
security_pgp_d	8336	8228 ↓ 1.30%	8312 ↓ 0.29%	7741 ↓ 7.14%
security_pgp_e	2971	3989 ↑ 34.26%	2904 ↓ 2.26%	2677 ↓ 9.90%
security_rijndael_d	13731	13332 ↓ 2.91%	13334 ↓ 2.89%	13144 ↓ 4.27%
security_rijndael_e	14636	12984 ↓ 11.29%	13569 ↓ 7.29%	13487 ↓ 7.85%
security_sha	3090	2804 ↓ 9.26%	3419 ↑ 10.65%	2902 ↓ 6.08%
telecom_CRC32	1221	823 ↓ 32.60%	968 ↓ 20.72%	961 ↓ 21.29%
telecom_adpcm_c	2918	3080 ↑ 5.55%	3210 ↑ 9.99%	2968 ↑ 1.71%
telecom_adpcm_d	2051	2113 ↑ 3.02%	2176 ↑ 6.09%	2064 ↑ 0.63%
telecom_gsm	2103	2382 ↑ 13.27%	1855 ↓ 11.79%	1807 ↓ 14.08%
Geomean (ms)	2442.49	2433.81 ↓ 0.36%	2544.65 ↑ 4.18%	**2325.83 ↓ 4.78%**

Table 3: Execution Time of PolyBench

Test case	LLVM-O3	Magellan	MLGO	LLM-Inliner
2mm	1096	980 ↓ 10.58%	990 ↓ 9.67%	1041 ↓ 5.02%
3mm	1577	1613 ↑ 2.28%	1605 ↑ 1.77%	1431 ↓ 9.26%
adi	6621	6725 ↑ 1.57%	6801 ↑ 2.72%	6507 ↓ 1.72%
atax	15	14 ↓ 6.67%	15 ↓ 0.00%	14 ↓ 6.67%
bicg	20	20 ↓ 0.00%	21 ↑ 5.00%	22 ↑ 10.00%
cholesky	19313	19876 ↑ 2.91%	19670 ↑ 1.85%	14678 ↓ 24.00%
correlation	5172	5326 ↑ 2.98%	5389 ↑ 4.20%	5141 ↓ 0.60%
covariance	5341	5201 ↓ 2.62%	5360 ↑ 0.36%	5197 ↓ 2.70%
deriche	210	207 ↓ 1.43%	203 ↓ 3.33%	212 ↑ 0.95%
doitgen	272	207 ↓ 23.90%	212 ↓ 22.06%	269 ↓ 1.10%
durbin	3	2 ↓ 33.33%	3 ↓ 0.00%	3 ↓ 0.00%
fdtd-2d	1092	1212 ↑ 10.99%	1193 ↑ 9.25%	1118 ↑ 2.38%
floyd-warshall	7896	7802 ↓ 1.19%	7722 ↓ 2.20%	7874 ↓ 0.28%
gemm	243	232 ↓ 4.53%	251 ↑ 3.29%	239 ↓ 1.65%
gemver	32	29 ↓ 9.38%	31 ↓ 3.13%	27 ↓ 15.63%
gesummv	14	13 ↓ 7.14%	13 ↓ 7.14%	14 ↓ 0.00%
gramschmidt	5048	5102 ↑ 1.07%	5094 ↑ 0.91%	3876 ↓ 23.22%
heat-3d	1442	1430 ↓ 0.83%	1387 ↓ 3.81%	1271 ↓ 11.86%
jacobi-1d	4	4 ↓ 0.00%	2 ↓ 50.00%	2 ↓ 50.00%
jacobi-2d	1091	1114 ↑ 2.11%	1083 ↓ 0.73%	986 ↓ 9.62%
lu	21753	21732 ↓ 0.10%	22527 ↑ 3.56%	21475 ↓ 1.28%
ludcmp	21535	21790 ↑ 1.18%	21437 ↓ 0.45%	21066 ↓ 2.18%
mvt	28	29 ↑ 3.57%	26 ↓ 7.14%	26 ↓ 7.14%
nussinov	2193	2098 ↓ 4.33%	2132 ↓ 2.78%	2001 ↓ 8.76%
seidel-2d	11587	11581 ↓ 0.05%	11552 ↓ 0.30%	11722 ↑ 1.16%
symm	1304	1265 ↓ 2.99%	1364 ↑ 4.60%	1254 ↓ 3.83%
syr2k	2750	2841 ↑ 3.31%	2838 ↑ 3.20%	2250 ↓ 18.18%
syrk	332	335 ↑ 0.90%	439 ↑ 32.23%	350 ↑ 5.42%
trisolv	9	9 ↓ 0.00%	9 ↓ 0.00%	8 ↓ 11.11%
trmm	1176	1193 ↑ 1.45%	1065 ↓ 9.44%	914 ↓ 22.28%
Geomean (ms)	566.43	550.04 ↓ 2.89%	551.81 ↓ 2.58%	**520.35 ↓ 8.14%**

Table 4: Performance of ChainMaker Dataset. The values in the table represent the geometric mean TPS of the dataset; bigger values indicate better inlining performance.

Application	Test Case	Go		Rust	
		Binaryen -O4	LLM-Inliner	LLVM -O3	LLM-Inliner
ERC721	mint	2472	2507 ↑ 1.42%	3951	4889 ↑ 23.74%
	setApprovalForAll	3757	4070 ↑ 8.33%	8548	8693 ↑ 1.70%
	transferFrom	2750	2869 ↑ 4.33%	2152	2268 ↑ 5.39%
exchange	buyNow	191	249 ↑ 30.37%	441	429 ↓ 2.72%
identity	addWriteList	654	655 ↑ 0.15%	797	924 ↑ 15.93%
	isApprovedUser	662	680 ↑ 2.72%	887	912 ↑ 2.82%
raffle	registRaffle	2322	2572 ↑ 10.77%	3948	4301 ↑ 8.94%
evidence	evidenceAndFind	3180	3103 ↓ 2.42%	2175	2667 ↑ 22.62%
itinerary	saveAndQueryHistory	1038	1193 ↑ 14.93%	1967	1958 ↓ 0.46%
fact	saveAndFind	2048	2092 ↑ 2.15%	3802	3681 ↓ 3.18%
compute	normalCal	288	293 ↑ 1.74%	9637	11112 ↑ 15.31%
	hashCal	140	454 ↑ 224.29%	170	174 ↑ 2.35%
	bigNumCal	33	46 ↑ 39.39%	40	42 ↑ 5.00%
	parseInt	188	236 ↑ 25.53%	9266	11168 ↑ 20.51%
Geomean (TPS)		710.13	**844.21 ↑ 18.88%**	1623.47	**1754.20 ↑ 8.05%**

Table 5: Real-world WebAssembly Dataset

Application	Test Case	Description
ERC721	mint	Mint assets to the designated account.
	setApprovalForAll	Grant the caller permission to manage assets of the target account.
	transferFrom	Transfer assets from one account to another.
identity	addWriteList	Batch-write 100 addresses into the whitelist.
	isApprovedUser	Batch-check whether 100 given addresses exist in the whitelist.
exchange	buyNow	Cross-contract invocation: calls functions from `identity` (`addWriteList`, `isApprovedUser`) and `ERC721` (`mint`, `setApprovalForAll`, `transferFrom`).
raffle	registRaffle	Register a raffle pool and execute the lottery logic.
evidence	evidenceAndFindByHash	Store evidence and retrieve records by hash value.
itinerary	saveAndQueryHistory	Save itinerary records and query historical itineraries.
fact	saveAndFindByFileHash	Simplified version of standard-evidence.
compute	normalCal	Perform 10^6 iterations of loop accumulation.
	hashCal	Execute SHA-256 hash computation for 10^4 times.
	bigNumCal	Conduct 10^4-times modular exponentiation on big integers.
	parseInt	Execute 3×10^4 times of string-to-integer base conversion.